

PRE-MEDICAL : ENTHUSIAST COURSE PHASE - ENTHUSIAST, LEADER & ACHIEVER COURSE

PHASE : ALL ENTHUSIAST, MLA, MAZA, MAZB, MAZC, MAZD, MAZL, MAZN, MAZO,

MAAX, MAAY, MAPA, MAPB, LAKSHYA

TARGET : PRE MEDICAL 2025

Test Type : **MAJOR**

Test Pattern : **NEET (UG)**

TEST DATE : 03-04-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	2	2	3	3	3	1	4	4	1	2	1	3	4	2	4	3	3	1	4	2	1	1	1	1	4	1	2	4	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	3	3	4	3	4	3	2	3	2	2	4	1	2	1	4	3	1	2	1	1	1	4	2	2	3	4	3	3	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	3	4	1	4	4	3	4	3	4	1	3	3	2	2	1	3	4	4	1	4	3	3	1	2	1	1	2	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	3	3	3	1	1	4	4	3	2	4	2	1	3	2	2	4	4	2	4	1	2	3	3	1	1	1	4	3	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	1	3	4	2	3	2	2	3	3	3	2	4	3	4	2	1	3	3	3	2	3	2	1	2	3	3	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	1	1	2	3	2	1	1	4	1	3	2	2	4	4	1	4	1	2	2	1	3	2	1	1	4	4	1	2	2

HINT - SHEET

6. **Ans (3)**

NCERT Pg. # 218

7. **Ans (1)**

Answer- Option(1)

Explanation - Let's analyze both the Assertion (A) and Reason (R) carefully.

Concept- Assertion (A): Ti^{3+} salts are colored. This is true. Ti^{3+} has an electronic configuration of $[Ar] 3d^1 4s^0$. The presence of an unpaired d electron allows for d-d transitions.

17. **Ans (3)**

NCERT-XII, Pg.No. 188

18. **Ans (3)**

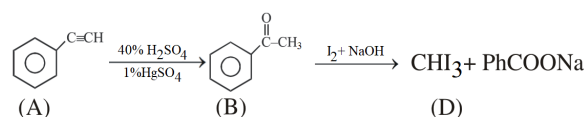
Inert pair effect.

19. **Ans (1)**

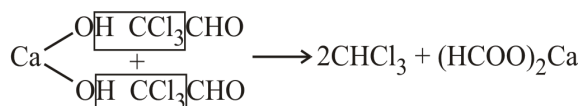
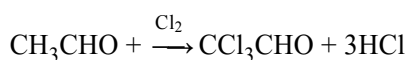
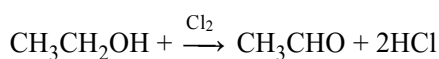
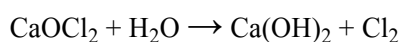
Order of ease of liquefaction :-

$Xe > Kr > Ar > Ne > He$

23. **Ans (1)**

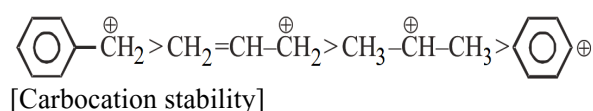


24. **Ans (1)**

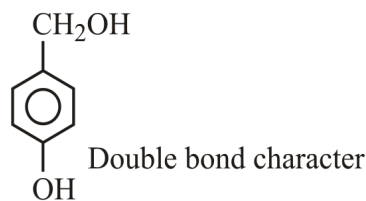


25. **Ans (1)**

Reactivity towards $S_N1 \propto$ stability of carbocation.



26. Ans (4)



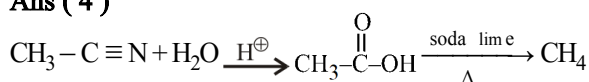
27. Ans (1)

O — m — p-dichlorobenzene
b.p.(K) 453 249 323

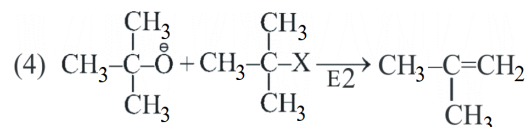
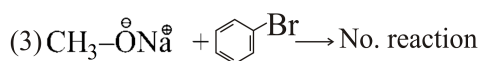
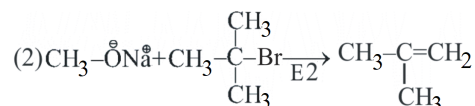
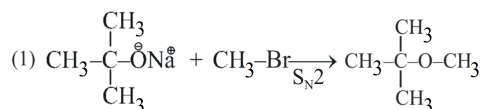
28. Ans (2)

It is Reimer Tiemann Reaction.

29. Ans (4)



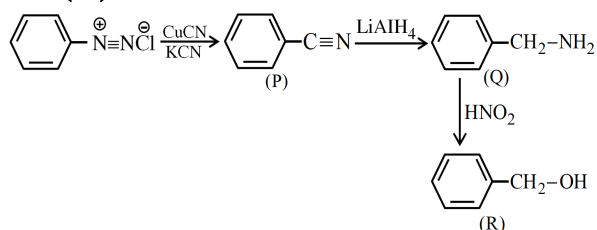
30. Ans (1)



31. Ans (3)

Y is $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$ which gives 2,4-D.N.P. test.

32. Ans (3)



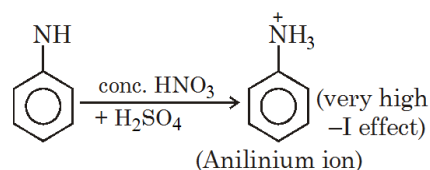
33. Ans (3)

1° amine in which NH_2 not attached to benzene can be prepared by Gabriel Phthdimide synthesis
 $\therefore$ Ethyl Amine ($\text{CH}_3\text{CH}_2\text{NH}_2$) can be prepared

34. Ans (4)

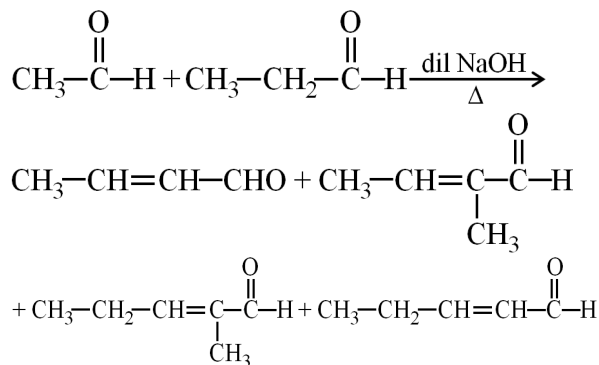
Propan - 1 - amine & Propan - 2 - amine, both are 1° Amine.
Hence Cannot be distinguished by Hinsberg reagent.

35. Ans (3)



Aniline on protonation gives anilinium ion which is meta directing. So considerable amount of meta product is formed.

36. Ans (4)



38. Ans (2)

Reactivity of carbonyl compounds

$$\propto \frac{-M, -H, -I}{+M, +H, +I}$$

39. Ans (3)

Disaccharides which donot have Hemi acetal group are non reducing sugar.

40. Ans (2)

Theory based NCERT Page No. 358

41. Ans (2)

$$\% \text{ Br} = \frac{80}{188} \times \frac{m}{w} \times 100;$$

$$= \frac{80}{188} \times \frac{0.94}{0.8} \times 100 = 50\%$$

42. Ans (4)

NCERT XI Pg. # 353

43. Ans (1)

$$\% \text{ of N} = \frac{1.4 \times M \times 2(V - V_1/2)}{m}$$

$$= \frac{1.4 \times 0.5 \times 2(50 - 60/2)}{0.5}$$

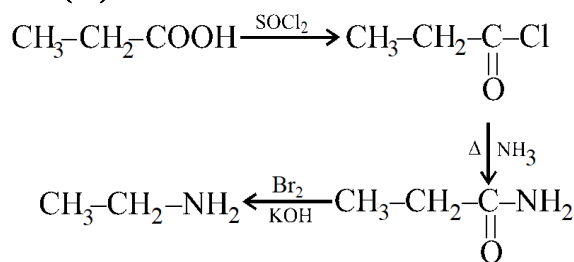
$$= 1.4 \times 40 = 56.0$$

M = molarity of H_2SO_4 , V = Volume of H_2SO_4

m = weight of sample of molarity M

V_1 = Volume of NaOH of molarity M

44. Ans (2)



45. Ans (1)

Rate of hydrolysis $\propto$ +ve charge at carbonyl carbon of Acid derivatives

46. Ans (4)

$$w = \frac{1}{2} I \omega^2 = MB [\cos \theta_1 - \cos \theta_2]; \theta_1 = 0^\circ, \theta_2 = 90^\circ$$

$$= \frac{1}{2} I \omega^2 = MB$$

$$\omega = \sqrt{\frac{2MB}{I}}$$

47. Ans (3)

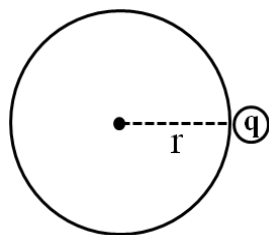
$$\tau = MB \sin \theta \quad (\theta = 90^\circ)$$

$$\tau = (IA) B$$

$$\tau = (I) \left[\frac{\pi R^2}{2} + \frac{4\pi R^2}{2} \right] B$$

$$\tau = \frac{5\pi R^2}{2} IB$$

48. Ans (1)



$$I = \frac{qu}{2\pi r}$$

$$B = \frac{\mu_0 I}{2\pi r}$$

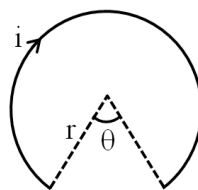
$$B = \frac{\mu_0}{2r} \frac{qu}{2\pi r}$$

$$q = \frac{4\pi r^2 B}{\mu_0 u}$$

50. Ans (1)

$$B = \frac{\mu_0 I}{4\pi R} (\theta)$$

51. Ans (1)



$$B = \frac{\mu_0 i}{4\pi R} (2\pi - \theta) \Rightarrow B = \frac{\mu_0 i (2\pi - \theta)^2}{4\pi (2\pi R)}$$

$$B = \frac{\mu_0 i}{2R} \left(\frac{2\pi - \theta}{2\pi} \right)^2 \because (2\pi - \theta)(r) = 2\pi R$$

$$r = \frac{2\pi R}{(2\pi - \theta)}$$

52. Ans (1)

$$B = \mu_0 n i$$

$$i' = (2i)$$

$$n' = n/2$$

$$B' = \mu_0 n' i' \Rightarrow B' = \mu_0 \frac{n}{2} (2i) \Rightarrow B' = B$$

53. Ans (4)

$$F = qvB \sin \theta \quad (\theta = 90^\circ)$$

$$F \propto \sqrt{v} \quad \frac{1}{2} m v^2 = qV$$

$$\frac{F'}{F} = \frac{\sqrt{5V}}{\sqrt{V}} \quad v \propto \sqrt{V}$$

$$F' = \sqrt{5} F$$

54. Ans (2)

$$B_1 = \frac{\mu_0 I}{2r}, M = IA = I(\pi r^2)$$

$$B_2 = \frac{\mu_0 I}{2\sqrt{2}r}, M' = 2M = IA' = I\pi(r')^2$$

$$B_2 = \frac{B_1}{\sqrt{2}}, r' = \sqrt{2}r$$

$$\frac{B_1}{B_2} = \frac{\sqrt{2}}{1}$$

55. Ans (2)

0 for all value of θ

56. Ans (3)

$$V = V_0 \sin 100\pi t$$

$$\omega = 100\pi$$

$$L = \frac{1}{\pi} mH = \frac{1}{\pi} \times 10^{-3} H$$

$$\omega L = 100\pi \times \frac{1}{\pi} \times 10^{-3} = 0.1 \Omega$$

$$I_0 = \frac{V_0}{\omega L} = \frac{10}{.1} = 100 A$$

$$I = 100 \sin \left(100\pi t - \frac{\pi}{2} \right)$$

57. Ans (4)

$$Z = R = 50 \Omega$$

$$I = \frac{V}{Z} = \frac{100}{50} = 2A$$

$$V_R = V = 100 V$$

58. Ans (3)

$$R = \frac{20}{\pi} \Omega$$

$$I_0 = \frac{V_0}{Z} = \frac{V_0}{\sqrt{\left(\frac{1}{2\pi fC}\right)^2 + R^2}}$$

$$\Rightarrow I_0 = \frac{250\sqrt{2}}{\sqrt{\left(\frac{1}{(2\pi \times 50 \times 500 \times 10^{-6})^2} + \left(\frac{20}{\pi}\right)^2}} = \frac{250\sqrt{2}}{20\sqrt{2}/\pi}$$

$$I_0 = 12.5 \pi \text{ amp.}$$

59. Ans (3)

$$\text{At A : } X_C > X_L$$

$$\text{At B : } X_C = X_L$$

$$\text{At C : } X_C < X_L$$

61. Ans (3)

$$d\phi_B = \vec{B} \cdot d\vec{A}$$

$$d\phi_B = B dA \cos 0^\circ$$

$$d\phi_B = \left(\frac{\mu_0 I}{2\pi x} \right) (b dx)$$

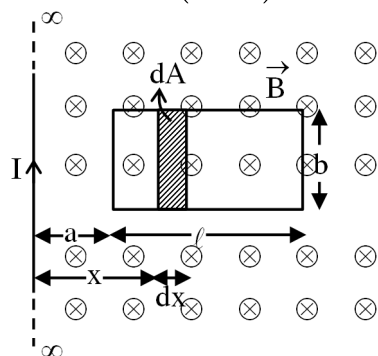
$$d\phi_B = \frac{\mu_0 I b}{2\pi x} dx$$

$$\int d\phi_B = \frac{\mu_0 I b}{2\pi} \int_a^{(a+\ell)} \frac{1}{x} dx$$

$$\phi_B = \frac{\mu_0 I b}{2\pi} [\ln(x)]_a^{(a+\ell)}$$

$$\phi_B = \frac{\mu_0 I b}{2\pi} [\ln(a+\ell) - \ln(a)]$$

$$\phi_B = \frac{\mu_0 I b}{2\pi} \ln \left(\frac{a+\ell}{a} \right)$$



62. Ans (3)

No relative motion between the coils, will result in no change in magnetic flux linked with coil-1 due to the magnetic field created by primary current in coil-1. So there will not be any induced current in coil-1. Hence, the deflection in galvanometer will be zero. (null)

63. Ans (3)

Instantaneous induced emf

$$\Rightarrow e_{\text{inst}} = \frac{-d\phi_B}{dt} = -(2t - 5) = 5 - 2t$$

$$\text{at } t = 2 \text{ second} \Rightarrow e_{\text{inst.}} = 5 - 2 \times 2 = 1 \text{ volt}$$

Average induced emf

(from $t = 0$ second to $t = 2$ second)

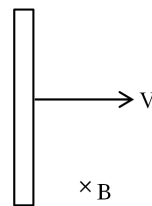
$$\Rightarrow e_{\text{avg.}} = \frac{-\Delta\phi_B}{\Delta t} = \frac{-(\phi_{B_2} - \phi_{B_1})}{(t_2 - t_1)}$$

$$\Rightarrow \frac{-[\{(2)^2 - 5(2) + 6\} - \{(0)^2 - 5(0) + 6\}]}{[2 - 0]}$$

$$\Rightarrow 3 \text{ volts}$$

$$\text{now ratio } \frac{e_{\text{inst.}}}{e_{\text{avg.}}} = \frac{1}{3}$$

64. Ans (4)



RHP Rule

$$E_{\text{end}} = 1 +ve$$

$$2 -ve$$

$$e = VB\ell$$

65. Ans (1)

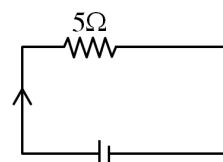
$$\vec{B} \parallel \vec{\ell}$$

Magnetic flux is not cut by rotational motion of rod so induced emf is zero.

$$(e = 0)$$

66. Ans (4)

Steady State



$$I = \frac{V}{R} = \frac{20}{5} = 4A$$

69. Ans (4)

$$\therefore i = \frac{10 - 0.7 - 0.3}{(1 + 1.5) \text{ k}\Omega} = \frac{9\text{V}}{2.5 \text{ k}\Omega} = 3.6 \text{ mA}$$

$$\therefore V_{R_L} = 3.6 \text{ mA} \times 1.5 \text{ k}\Omega = 5.4 \text{ volt}$$

70. Ans (3)

$\therefore$ Intrinsic semiconductor - Pure ($n_e = n_h$)

p-type semiconductor - $n_h \gg n_e$ (holes majority)

n-type semiconductor - $n_e \gg n_h$ (electron majority)

71. Ans (4)

From ckt, current in $1 \text{ k}\Omega$ resistor is

$$I = \frac{20 - 12}{R} = \frac{8}{1 \text{ k}\Omega} = 8 \text{ mA}$$

Power = $I V$

$$= (8 \times 10^{-3}) \times 8$$

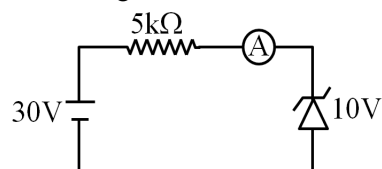
$$= 64 \text{ mW}$$

72. Ans (1)

In filter circuit capacitor connected in parallel and inductor connected in series with load resistance.

73. Ans (3)

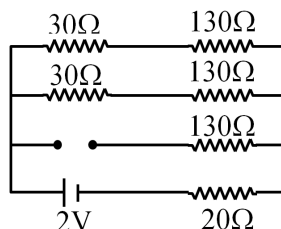
Here 2nd zener diode is in F.B. so diode replace by connecting wire.



$$I = \frac{30 - 10}{5 \text{ k}\Omega} = 4 \text{ mA}$$

74. Ans (3)

Equivalent circuit



$$R_{eq} = 80 + 20 = 100 \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{2}{100} = 20 \text{ mA}$$

75. Ans (2)

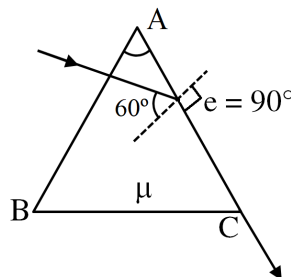
$$Y = \bar{A} + \bar{B} = \bar{A \cdot B} = A \cdot B = \text{AND gate}$$

78. Ans (3)

$$M.P. = \frac{v_0}{u_0} \times \frac{D}{f_e}$$

$$400 = \frac{20}{0.5} \times \frac{25}{f_e} \Rightarrow f_e = 2.5 \text{ cm}$$

79. Ans (4)



$$i = 0, A = 60^\circ, e = 90^\circ$$

$$\text{if } i = 0 \text{ then } r_1 = 0$$

$$A = r_1 + r_2$$

$$A = r_2 = 60^\circ$$

$$\mu \times \sin r_2 = 1 \times \sin e$$

$$\mu = \frac{\sin e}{\sin r_2} = \frac{\sin 90^\circ}{\sin 60^\circ} = \frac{2}{\sqrt{3}}$$

80. Ans (4)

Using the mirror formula :

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Here, $u = -15 \text{ cm}$, $v = -10 \text{ cm}$

$$\frac{1}{f} = \frac{1}{-15} + \frac{1}{-10} = \frac{-2-3}{30} = \frac{-5}{30}$$

$$f = -6 \text{ cm}$$

(Negative sign indicates the concave mirror)

82. Ans (4)

Path difference for 5th order minima

$$= (2n - 1) \frac{\lambda}{2}$$

$$= (2 \times 5 - 1) \frac{\lambda}{2} = \frac{9\lambda}{2}$$

84. Ans (3)

Say intensity of unpolarized light is I_0 .

Then,

$$\text{output from polariser} = \frac{I_0}{2}$$

Output from analyser

$$= \frac{I_0}{2} \cos^2 45^\circ = \frac{I_0}{4} = I \text{ (given)}$$

Therefore,

$$I_0 = 4I$$

86. **Ans (2)**

The number of photons emitted per second is

$$n = \frac{P}{h\nu} = \frac{3.315 \times 10^{-3}}{6.63 \times 10^{-34} \times 2.5 \times 10^{14}} \\ = 2 \times 10^{16}$$

NCERT-XII, Pg. # 284, Part-2

87. **Ans (1)**

$$\vec{F} = qE \Rightarrow a = \frac{qE}{m}$$

$$v = u + at$$

$$\text{For proton, } v = 0 + \frac{eEt}{m} \Rightarrow \lambda_p = \frac{h}{mv} = \frac{h}{eEt}$$

$$\text{For alpha, } v = 0 + \frac{2eEt}{m} \Rightarrow \lambda_\alpha = \frac{h}{mv} = \frac{h}{2eEt}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{2}{1}$$

88. **Ans (1)**

$$E = -E_0 \hat{i}$$

$$\leftarrow \quad \quad \quad e^- \rightarrow V_0$$

$$\leftarrow$$

$$\leftarrow$$

$$a = \frac{eE_0}{m}$$

$$v_f = u + at$$

$$v_f = v_0 + \frac{eE_0}{m}t$$

$$\lambda_f = \frac{h}{mv_0 \left(1 + \frac{eE_0}{mv_0}t\right)}$$

$$\lambda_i = \frac{h}{mv_0}$$

$$\lambda_f = \frac{\lambda_i}{2} \text{ (given)}$$

$$\text{So, } \frac{h}{2mv_0} = \frac{h}{mv_0 \left(1 + \frac{eE_0}{mv_0}t\right)}$$

$$1 + \frac{eE_0}{mv_0}t = 2 \Rightarrow t = \frac{mv_0}{eE_0}$$

90. **Ans (3)**

$$\text{From PEE equation, } eV_0 = h\nu - \phi_0 = \frac{hc}{\lambda} - \phi_0$$

$$eV_0 = \frac{12400}{1000} - 2.4$$

$$= 12.4 - 2.4 = 10.0 \text{ eV}$$

$$V_0 = 10.0 \text{ volt}$$

91. **Ans (4)**

NCERT Pg. # 67

92. **Ans (3)**

NCERT-XII, Page No. # 81, 88

93. **Ans (3)**

NCERT XII Pg # 94,95

94. **Ans (3)**

NCERT XII Page-No.78

95. **Ans (1)**

NCERT (XII) Page No. # 77

96. **Ans (1)**

XII NCERT Page No. # 101

97. **Ans (4)**

NCERT-XII, Pg. # 91

98. **Ans (4)**

NCERT XII Page No. 96

99. **Ans (3)**

NCERT Pg. # 95

100. **Ans (2)**

NCERT Pg. # 88

101. **Ans (4)**

NCERT XII, Pg. # 98(E), 109(H)

102. **Ans (2)**

NCERT Pg. # 106

103. **Ans (1)**

NCERT-XII, Pg # 92

104. **Ans (3)**

NCERT XII Pg. # 74

105. **Ans (2)**

NCERT Pg. No. 93

106. **Ans (2)**

NCERT XII Pg. # 67

107. **Ans (4)**

NCERT XII Page No. 70

108. **Ans (4)**

NCERT Pg. # 63

109. **Ans (2)**

NCERT, Pg. # 67-68

- | | |
|--|--|
| <p>110. Ans (4)
NCERT Pg. # 60</p> <p>111. Ans (1)
NCERT Pg. No. # 91</p> <p>112. Ans (2)
NCERT Pg. No. # 62, 69</p> <p>113. Ans (3)
NCERT, Pg # 153</p> <p>114. Ans (3)
NCERT, Pg. # 151, 153, 154, 155</p> <p>115. Ans (1)
NCERT, Pg. # 156, 157, 158</p> <p>116. Ans (1)
NCERT XII Pg. # 200</p> <p>117. Ans (1)
NCERT XII Pg. # 197, 199</p> <p>118. Ans (4)
NCERT, Pg. # 200</p> <p>119. Ans (3)
NCERT, Pg. # 197</p> <p>120. Ans (2)
NCERT XII Pg # 201</p> <p>121. Ans (3)
NCERT-XII, Pg. No. # 191</p> <p>122. Ans (3)
NCERT XII, Page No. 195</p> <p>123. Ans (1)
NCERT-XII, Pg. # 193</p> <p>124. Ans (3)
NCERT XII, Page No. 210</p> <p>125. Ans (4)
NCERT-XII, Pg. No. # 207</p> <p>126. Ans (2)
NCERT XII Pg. # 209(E)/228(H) Para : 12.4</p> <p>127. Ans (3)
NCERT Pg. # 211</p> | <p>128. Ans (2)
Module</p> <p>129. Ans (2)
NCERT Pg. No. # 210 (E), 229 (H)</p> <p>130. Ans (3)
NCERT, Pg # 224 (E), 245 (H)</p> <p>131. Ans (3)
NCERT XII Pg. # 221 (E), 241 (H)</p> <p>132. Ans (3)
NCERT, Pg # 220 (E), 240 (H)</p> <p>133. Ans (2)
NCERT Pg. # 223(E), 243(H)</p> <p>134. Ans (4)
NCERT Pg. # 224(E), 245(H)</p> <p>135. Ans (3)
NCERT Pg. # 225(E), 245(H)</p> <p>136. Ans (4)
NCERT Pg # 147</p> <p>137. Ans (2)
NCERT Pg.# 163</p> <p>138. Ans (1)
NCERT XII, Pg. # 196</p> <p>139. Ans (3)
NCERT XII Page No. # 195</p> <p>140. Ans (3)
NCERT XII, Pg. # 199</p> <p>141. Ans (3)
NCERT Pg.# 164, 170</p> <p>142. Ans (2)
NCERT XII Page No. # 170</p> <p>143. Ans (3)
NCERT Pg # 152</p> <p>144. Ans (2)
NCERT Pg. # 167, 169</p> <p>145. Ans (1)
NCERT, Pg. # 174</p> |
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146. **Ans (2)**
NCERT-XII, Pg. # 208
147. **Ans (3)**
NCERT XII Page-No-208
148. **Ans (3)**
NCERT, Pg. # 208
149. **Ans (2)**
NCERT XII Pg.-211
150. **Ans (3)**
NCERT-XII, Pg. # 183 (E), 200 (H)
151. **Ans (1)**
NCERT Pg # 213 (E), 201-202 (H)
152. **Ans (1)**
NCERT XII Pg.-213
153. **Ans (1)**
NCERT XII, Pg. # 214 & 215
154. **Ans (2)**
NCERT, Pg. # 230
155. **Ans (3)**
NCERT Pg. # 151
156. **Ans (2)**
NCERT Pg. # 152 (E), 167 (H)
157. **Ans (1)**
NCERT XII, Pg. # 155
158. **Ans (1)**
NCERT-XII, Pg # 186
159. **Ans (4)**
NCERT Pg. # 137
160. **Ans (1)**
NCERT Pg. No. # 158, 159
161. **Ans (3)**
NCERT Pg. No. 151
162. **Ans (2)**
NCERT XII, Page # 133
163. **Ans (2)**
NCERT Pg. # 137
164. **Ans (4)**
NCERT (XIIth) Pg. # 160 (Para 8.5)

165. **Ans (4)**
NCERT XII Pg.#162,163 (E) 176(H)
166. **Ans (1)**
NCERT-XII, Page No. 152 (E) Para = 8.2.3 and 164 (H)
Some maternal antibodies (preformed antibodies) like Ig-G reaches in foetus through placenta provides passive immunity to the foetus.
167. **Ans (4)**
NCERT Page No. 134
168. **Ans (1)**
NCERT Pg. # 142
169. **Ans (2)**
NCERT Pg.# 147
170. **Ans (2)**
NCERT Pg. # 122
171. **Ans (1)**
NCERT- P.no. 118
Bobcat is placental mammal.
172. **Ans (3)**
NCERT- Pg.No. 111, 118, 119
Work of Thomas Malthus on population influenced Darwin.
173. **Ans (2)**
NCERT Pg. # 122
174. **Ans (1)**
NCERT Pg # 114, 126
175. **Ans (1)**
NCERT Pg # 122
176. **Ans (4)**
NCERT(XII) Pg#138/148(H) Para:7.8
177. **Ans (4)**
NCERT (XII) Pg. # 133
178. **Ans (1)**
NCERT XII Pg. No. # 116
179. **Ans (2)**
NCERT(XII) Pg#140/151(H) Para:7.9
180. **Ans (2)**
NCERT-XI Pg. # 114-115